**Morgan Rivera:** mfrivera_scoring_eda_.pynb

To determine the strongest offensive team, I focused on scoring, which plays a crucial role in identifying a winning team. This analysis examines how effectively teams capitalize on their scoring opportunities by evaluating key batting statistics. Specifically, I analyzed RBI, HR, and AVG. Batting Average (AVG): Indicates how often a team gets on base and successfully makes a hit. Home Runs (HR): Highlights a team's power-hitting ability. Runs Batted In (RBI): An indicator of effectiveness in both home runs and other scoring plays.

The objective is to identify the top-performing teams based on batting statistics and assess which teams maximize their scoring potential. The unit of analysis is the teams. The process involves aggregating team statistics to compute totals for RBI and HR while averaging AVG. Filtering teams above the average in all three metrics to ensure we analyze well-rounded, high-performing teams. Normalizing the data brings different scales (HR: 300-600, RBI: 1200-2000, AVG: 0.2-0.3) into comparable ranges. Visualizing the final set of teams using bar plots to compare performance effectively. By analyzing these metrics together, we can identify the most effective offensive teams.

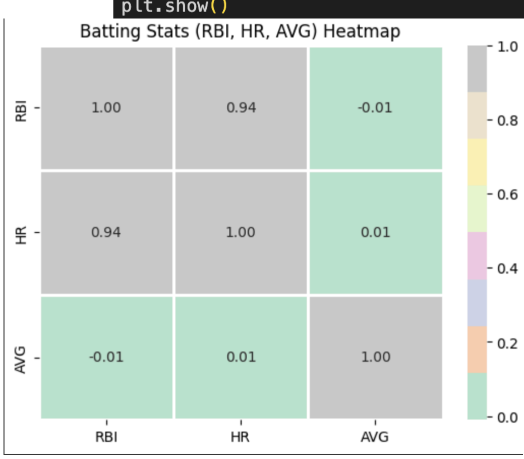
This query focused on finding well-rounded teams that are strong scorers. The steps were grouping data by team, summing up their total RBI and HR, and averaging their AVG. Computing the MLB average for RBI, HR, and AVG. Filtering teams that performed above average in all three categories, ensuring a balanced comparison. The query returned 8 rows in 0.0305 seconds. First 2 rows of query below.

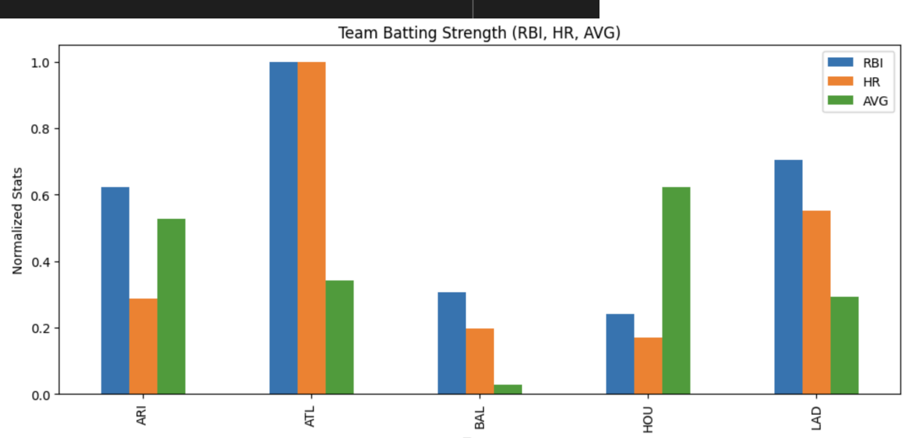
| Team | RBI | HR | AVG |
|---|---|---|---|
| ARI | 1759 | 448 | 0.250571 |
| ATL | 1991 | 629 | 0.247722 |

The heatmap shows the correlation, highlighting a strong correlation of 0.94 between HR and RBI. This indicates that as a team's RBI increases, its HR tends to increase as well. There is no correlation between AVG and RBI or AVG and HR, meaning AVG does not affect these. For the bar chart, the normalized scale of the data allows for a fair comparison of offensive statistics. This shows how RBI, HR, and AVG align on the same scale. Some teams like ARI and HOU with a higher batting average (AVG) may be making frequent hits but not translating them into RBI or HR effectively. Teams like ATL, BAL, and LAD with high RBI and HR, even with a lower AVG, show strong offensive production despite making fewer total hits. By visualizing team performances, we can identify the best offensive teams and understand how their strengths contribute to overall scoring success.

```python
# heat map of correlation between RBI, HR, and AVG
sns.heatmap(team_stats[['RBI', 'HR', 'AVG']].corr(), annot=True, fmt='.2f', cmap='Pastel2', linewidths=2)
plt.title('Batting Stats (RBI, HR, AVG) Heatmap')
plt.show()
```

```python
# compare top 5 teams with RBI >= .24 with barplot displaying RBI, HR, AVG
# normalize stats so all values range from 0-1 and can be compared
normalized_stats = (team_stats - team_stats.min()) / (team_stats.max() - team_stats.min())

# filter stats to focus only on teams that have an RBI >= .24
filtered_stats = normalized_stats[normalized_stats["RBI"] >= 0.24]
filtered_stats.plot(kind="bar", figsize=(12, 5))
plt.title("Team Batting Strength (RBI, HR, AVG)")
plt.ylabel("Normalized Stats")
plt.xticks(rotation=90)
plt.show()
```

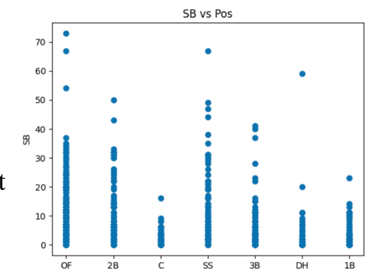


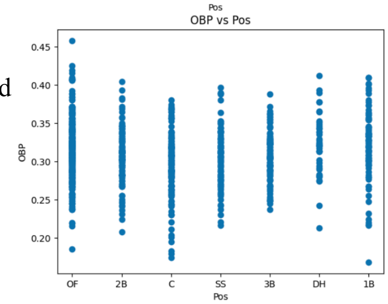
**David Sweasey:** hw6_eda_david_sweasey

I have determined a subgoal of my exploratory data analysis to find which teams have the fastest players. Speed is important on offense because it allows players to reach base more often on close calls, and helps advance themselves into scoring position. In most cases, a single hit can send someone from scoring position to home which results in a point: the whole goal of offense. Therefore, speed is a crucial factor in a team's offensive abilities.

I analyzed two key metrics that can be indicators of player speed. First, **OBP** (on-base percentage) depicts how many times a player makes it on base. This includes walks and hit by pitch, which means that OBP is not a perfect metric, but we can assume that a higher OBP means a player tends to be faster. Second, **SB** (stolen bases), which is how many times a player advances the bases without another player hitting the ball. To successfully steal a base, you need to be fast. If not, once the catcher receives the ball from the pitcher, he will attempt to throw the ball to the base to get you out. Stolen bases are therefore a better metric for speed.




I briefly analyzed how these two metrics compared to a player's **Pos** (position). Typically, positions like shortstop and outfield require faster players to make plays, whereas catchers and first-basemen are not required to be as mobile. However, most of the time a team is not going to benefit from having 4 shortstops, since only 1 can play at a time, so I did not continue with this analysis. We can see below, however, that the "fast" positions tend to have more stolen bases.

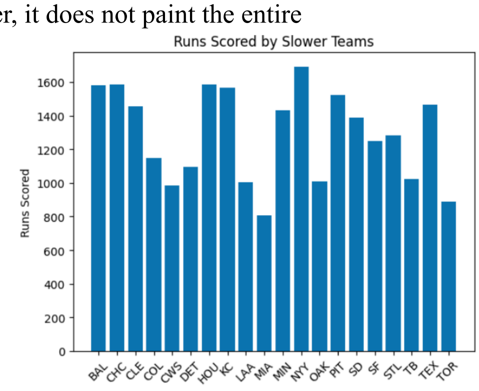


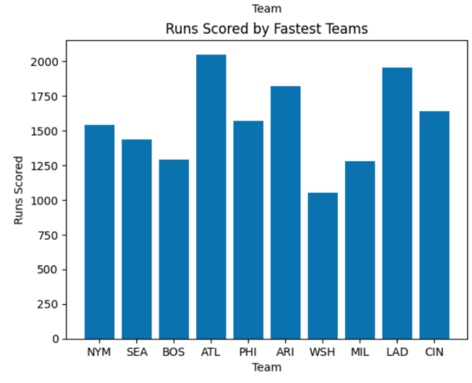

The query I performed was to find the top 10 fastest teams by the two aforementioned metrics. First, I had to aggregate them by team. I summed all stolen bases, and averaged all on base percentages. Then, I dropped the teams in the lower 50% of OBP. Although OBP is not a perfect metric as I explained, a team in the bottom 50% is still very likely to be on the slower side. Additionally, a lower OBP will tend towards less stolen bases. Then, I drop teams with stolen bases not in the top 10 of remaining teams. This gives the following which ran in 0.016 seconds:

| | Team | OBP | SB |
|---|---|---|---|
| 0 | CIN | 0.315514 | 378 |
| 1 | LAD | 0.323688 | 323 |

I want to see if being a fast team means you score more points. I plotted runs scored for each of these teams. The range was from just over 1000 runs to over 2000 runs, which is quite large. Then, I created a dataset with the teams that were not included in the fastest teams to see how many runs they scored. The following makes it clear that faster teams do tend to score more runs, so for a team to have a great offensive season, having fast players is a large benefit. However, it does not paint the entire picture, because some teams with the most runs are not the fastest.

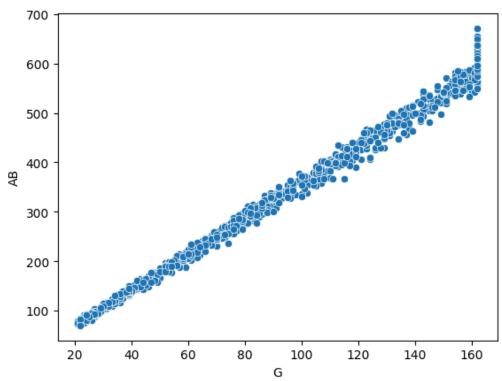
```python
# Check how many runs each of the fastest teams scored

# First, only keep the teams from the original dataset that are in the speed_stats
fastest_teams = df[df['Team'].isin(speed_stats['Team'])]
slower_teams = df[~df['Team'].isin(speed_stats['Team'])]
# Now, group by team and sum the runs scored
runs_scored = fastest_teams.groupby('Team')['R'].sum().reset_index()
runs_scored_slow = slower_teams.groupby('Team')['R'].sum().reset_index()

# Merge with fastest teams
runs_scored = pd.merge(runs_scored, speed_stats, on='Team').sort_values(by='SB', ascending=True).reset_index(drop=True)

# Display barchart of runs scored in order of fastest teams
plt.bar(runs_scored['Team'], runs_scored['R'])
plt.title('Runs Scored by Fastest Teams')
plt.xlabel('Team')
plt.ylabel('Runs Scored')
plt.show()

# Display barchart of the runs scored of slower teams
plt.bar(runs_scored_slow['Team'], runs_scored_slow['R'])
plt.title('Runs Scored by Slower Teams')
plt.xlabel('Team')
plt.xticks(rotation=45)
plt.ylabel('Runs Scored')
plt.show()
```



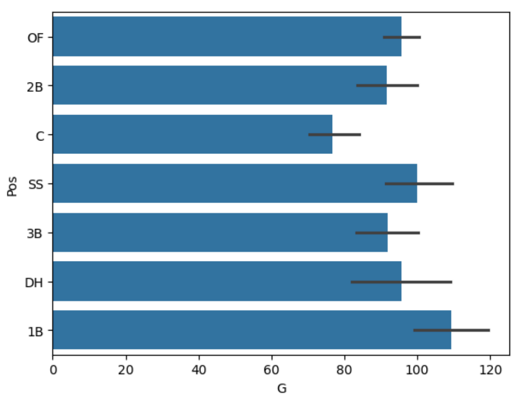



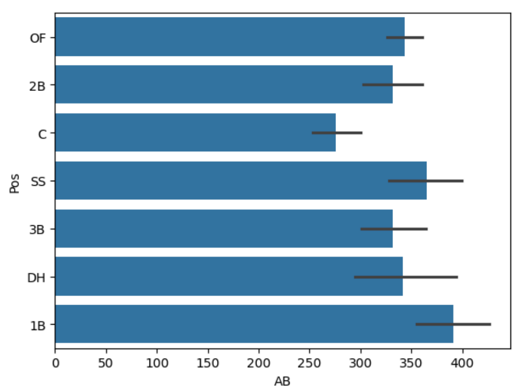
**John Farrell:** relationship between games, position and at bat.ipynb

To determine which positions the most impactful players play while on defense, I looked at the relationship between games, at bats, and position. Games is the number of games played, at bats is the number of times a player swung a bat at plate, and position is the position the player plays on defense. These data points are being analyzed together in order to find the most impactful position in baseball and find which team has the most impactful position players in their team.




The first relationship I investigated was between games and at bats which formed a linear positive correlation. This relationship is likely because the average player gets a similar number of bats each game, with the only outliers in this graph being at the top of the chart with important players who are at bat higher than average and as such are put in more games.




The second relationship I looked into was the relationship between the position of the player and the number of games played and number of at bats. Both charts look nearly identical, likely because of the relationship between at bats and games. Both graphs also show a most impactful position on a team being first baseman, as such a team with more first baseman is more likely to perform well offensively

The query I performed was to find the average number of players of each position. First I found the unique number of teams and positions. Then for each position I found the number of players in that position in the dataset and divided by the number of teams. This process took .006 seconds and was 7 lines long.

```
OF 10.766666666666667
2B 3.9
C 4.466666666666667
```

## Analysis Sketch

Our goal is to predict the 2025 MLB World Series Champion. To do this, we are examining the MLB offensive player statistics from the 2023 and 2024 seasons. Offense is what scores points, and you cannot win a World Series without scoring. We chose the team as the unit of analysis, and much of our analysis revolves around aggregating these statistics into total team statistics. Morgan specifically focused on scoring performance by analyzing RBI, HR, and AVG. This approach provides insight into well-rounded offensive powerhouse teams that get on base and put points on the scoreboard. David explored the relationship between team speed and runs scored. He revealed that speed is important on offense, which can result in more runs than the average team. John analyzed the importance of the positions and found that first basemen tend to be the most important to an offense. They also may be less injury prone, so a team with a strong first baseman might be more successful.

During our analysis, we plan to train a decision tree using the statistics that seem to be most important from our independent exploratory data analysis. These statistics are **RBI/HR,** where RBI/HR is a good indicator of a strong hitting team, **SB**, which is the best offensive indicator of a team's speed, and **R,** which is a great metric for offensive power because it simply is how many points a team scored. Since our unit of analysis will be the team, our decision tree will be trained and tested on aggregated team stats, similar to ones we used in the above explorations.

To ensure we are including star players that tend to make a large difference in the outcome of games, instead of focusing on only team averages, we are going to incorporate some of John's analysis and calculate some sort of offensive skill metric for each player. We will take the highest two or three (not sure yet) players and add them to a dataframe of team statistics so that the decision tree can also use these select players to help with prediction. We are unsure of exactly how to calculate this metric. David will look into this in the next couple of days.

The classes for the decision tree will be a range of team wins. The first class will be the range of wins held by the lowest 6 teams, the second will be the range of the next 6 teams, and so on. We will train our decision tree to predict which class, or bucket of wins, each time will fall in. As we calculate training and testing accuracy, we may adjust our model or the number of classes to improve this accuracy. Then, we can use our model on a current team's roster to predict how many wins they should get, which will get us closer to figuring out who is winning this year's World Series!

**Responsibilities for Analysis**

| Morgan | David | John |
|---|---|---|
| Find Top Players - **4/4** | Create Offensive Skill Metric - **4/2** | Find W/L for 2024 Teams - **4/4** |
| Revise Decision Tree - **4/11** | Create Decision Tree - **4/6** | Revise Decision Tree - **4/11** |
| Final Analysis - **4/13** | Final Analysis - **4/13** | Final Analysis - **4/13** |
| Poster and Presentation - **4/14** | Poster and Presentation - **4/14** | Poster and Presentation - **4/14** |